

# FF-1 EMBER

Full-size build plans, tiled for A4

## PRINT AT 100 % - ACTUAL SIZE

In the print dialog set Scale to 100 %. Turn OFF "Fit to page", "Shrink oversized pages" and "Scale to fit". Those options take 4-6 % off, which moves the centre of gravity and will make the aircraft unflyable.

050 mm100 mm

MEASURE THIS BAR - IT MUST BE 100 mm

25 mm square

## WHAT THIS IS

A foam prototype of the FF-1's aerodynamic shape: wing, tail and fuselage pod at exact full size. It is phase 1 of the flight test plan - the aircraft that gets you a measured sink polar before you commit to composite moulds.

It is NOT the solar aircraft. The array, the energy-recovery turbine and the retractable motor pylon all need the moulded airframe: foam has neither the stiffness for a flat cell array nor the internal volume for the mast trunk. Build this, fly it, log it, then cut moulds.

## KEY NUMBERS

|                              |                                                              |
|------------------------------|--------------------------------------------------------------|
| Span                         | 2000 mm                                                      |
| Wing area                    | 3580 cm2                                                     |
| Root chord / tip chord       | 195 / 92 mm                                                  |
| Aerofoil                     | FF-SC1, 9.5 % thick                                          |
| Wing incidence / V-tail      | +1.5 / -1.0 deg                                              |
| Dihedral, outer panels       | 4.5 deg from y=450 mm                                        |
| Washout at tip               | 2.0 deg                                                      |
| <b>CENTRE OF GRAVITY</b>     | <b>277 mm from nose = 62 mm aft of the wing leading edge</b> |
| Target all-up mass           | 833 g                                                        |
| Predicted stall / best glide | 5.6 / 6.5 m/s                                                |
| Predicted L/D / min sink     | 15.6 / 0.41 m/s                                              |

## SHEET INDEX

|       |                    |
|-------|--------------------|
| 3-9   | WING PANEL         |
| 10-11 | AEROFOIL TEMPLATES |
| 12-17 | FUSELAGE POD       |
| 18-20 | FORMERS            |
| 21-22 | V-TAIL PANEL       |

# ASSEMBLING THE SHEETS, AND BUILDING FROM THEM

## 1 TAPE THE TILES

Each tile overlaps the next by 12 mm. Trim along the dashed line on the overlapping sheet only, lay it over its neighbour and line up the solid triangles, then tape. Work along a row first, then join rows.

The corner cross marks should coincide when you are right.

## 2 TWO WAYS TO MAKE THE WING

**HOT-WIRE CORES** (accurate, needs a wire bow). Cut the aerofoil templates from 1.5 mm ply or aluminium, pin them to the ends of an XPS block at the right stations, and wire between them. This gives the true FF-SC1 section and is what the aerofoil sheets are for.

**FLAT FOAM** (quick, no tools). Cut the planform from 6 mm depron, score the underside along the spar line, and fold a thin leading-edge wrap over a carbon tube. You lose about 15 % of L/D against the true section but it flies, and the planform and CG are unchanged.

## 3 BUILD ORDER

Wing first, and weigh every part as you go - the mass budget on the title sheet is what the predicted performance assumes.

- Cut both wing panels. Mirror the template for the left.
- Bury the 8 mm carbon spar at 30 % chord, full span, and sleeve the joint at the centre.
- Set the outer panels up 4.5 deg from the panel break. Block the tip 2 deg trailing-edge-up for washout while the glue sets - without it the tips stall first and it will drop a wing.
- Skin, then cut the aileron and flap free along the hinge lines.
- Pod sides, formers, boom. Set the wing seat at +1.5 deg and the V-tail at -1.0 deg. That 2.5 deg decalage is the trim.
- Fit the gear, then **BALANCE** before the first flight.

## 4 BALANCE - DO NOT SKIP THIS

Centre of gravity 277 mm from the nose, which is 62 mm aft of the wing leading edge at the root. Marked on the wing and fuselage sheets.

That is 34 % of mean chord, giving 20 % static margin. Forward limit 26 %, aft limit 42 % of mean chord. Nose-heavy is survivable; tail-heavy is not. Move the battery to trim, and only add lead once the battery is against a stop.

## 5 WHAT TO EXPECT

At 833 g the wing loading is 23.3 g/dm<sup>2</sup> - lighter than the composite aircraft, so it flies slower and thermals better.

stall about 5.6 m/s, best glide 6.5 m/s, L/D around 16, minimum sink 0.11 m/s at 6.4 m/s

Hand launch into wind, wings level, firm push. Trim for a flat glide before touching the throttle.

## 6 THEN MEASURE IT

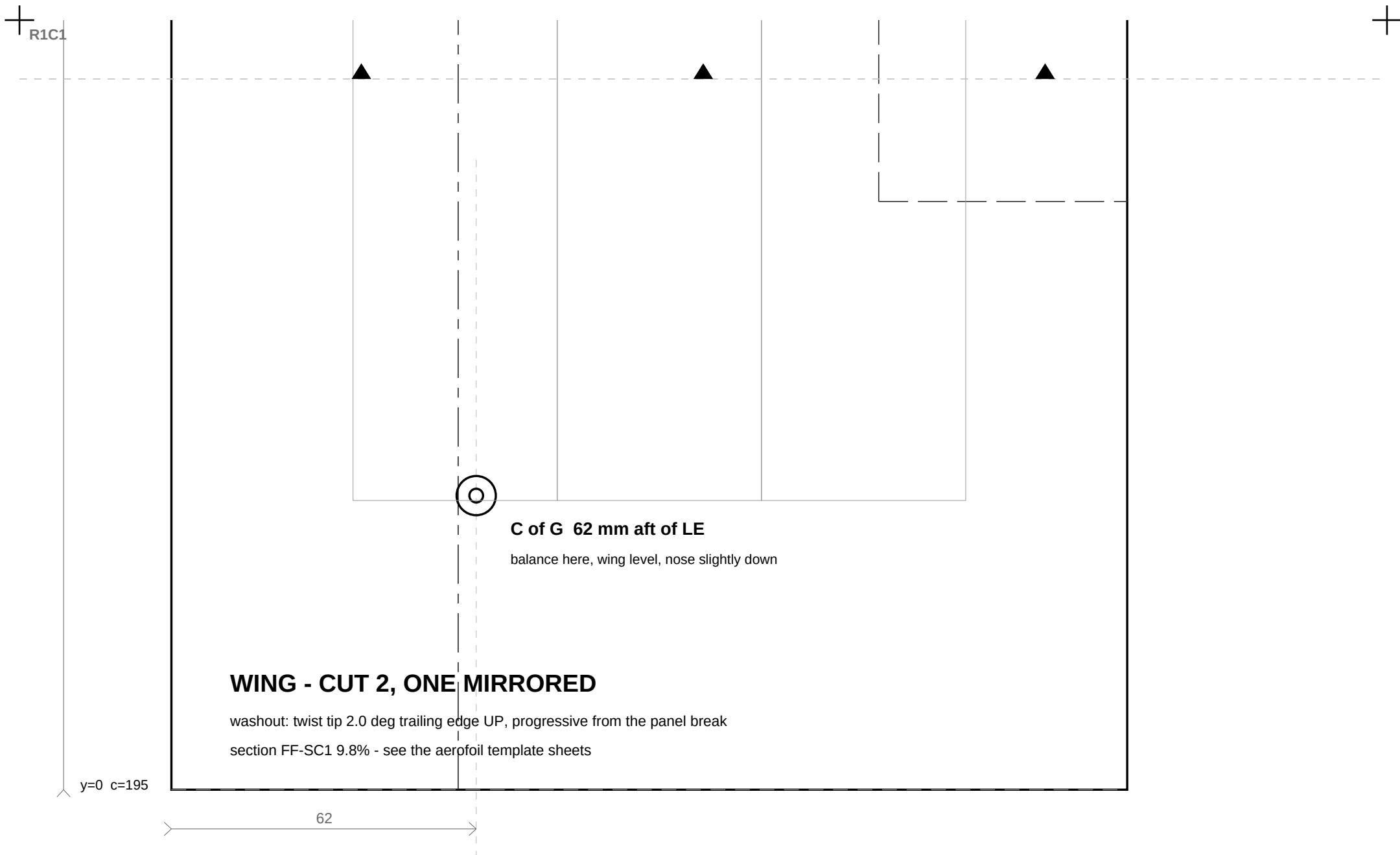
Fit the flight controller and an airspeed sensor, fly the polar sortie in docs/FLIGHT\_ANALYSIS.md section 4, and run:

```
python3 tools/flight_analysis.py YOURLOG.BIN --mass 0.80
```

# BUILD NOTES

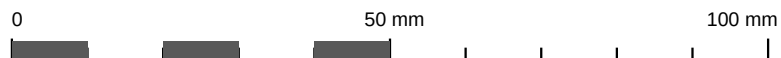
0

CHECK BEFORE



## WING PANEL

tile row 1 of 7, column 1 of 1 - overlap 12 mm, align the triangles



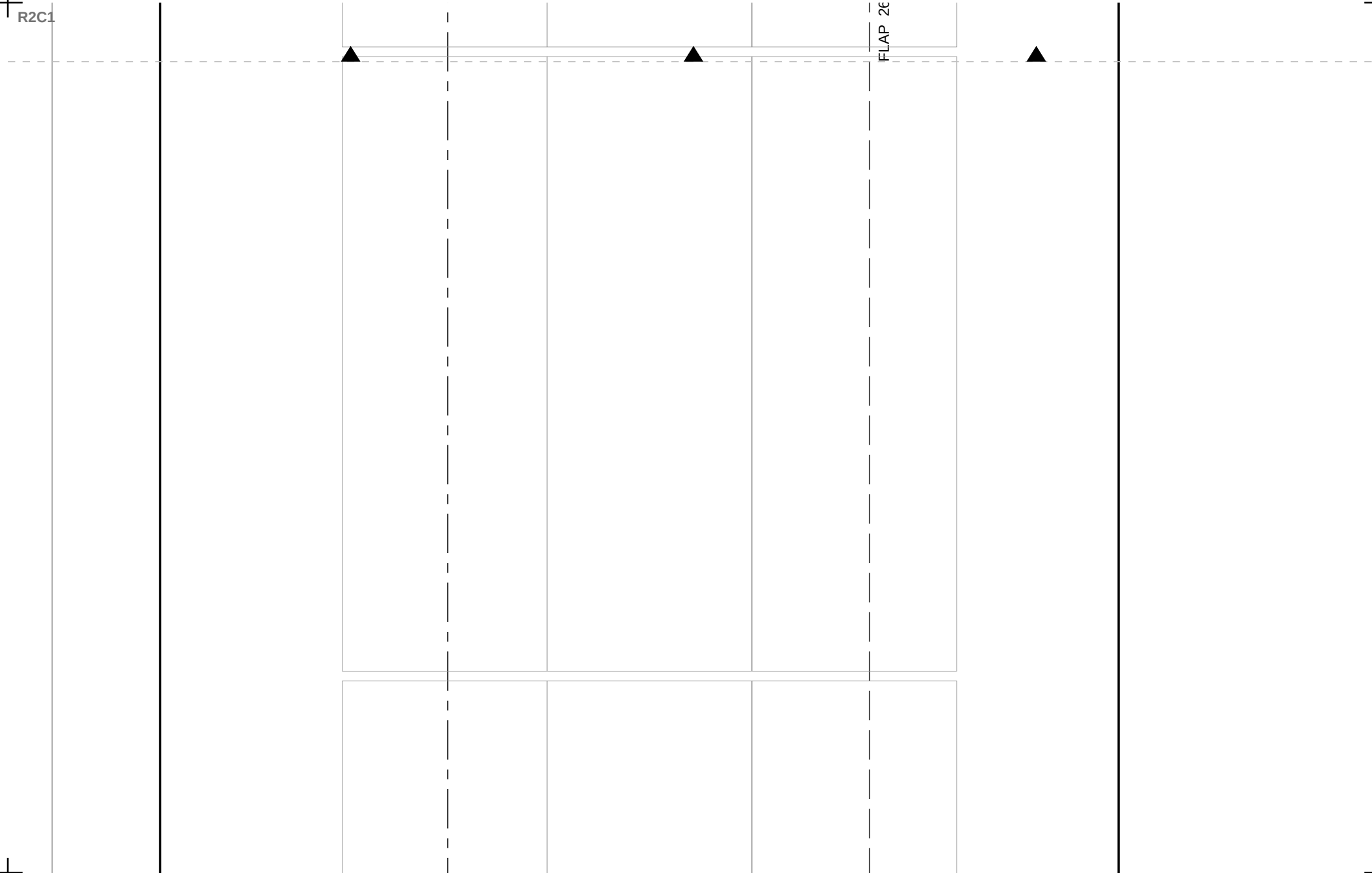
CHECK BEFORE CUTTING

## SHEET 3 / 22

FF-1 Ember - print at 100%



R2C1



**WING PANEL**

tile row 2 of 7, column 1 of 1 - overlap 12 mm, align the triangles

0

50 mm

100 mm

CHECK BEFORE CUTTING





R3C1



y=450 c=195

PANEL BREAK y=450 oui

FLAP 26% chord

## WING PANEL

tile row 3 of 7, column 1 of 1 - overlap 12 mm, align the triangles

0

50 mm

100 mm

CHECK BEFORE CUTTING

SHEET 5 / 22

FF-1 Ember - print at 100%

$+$ 

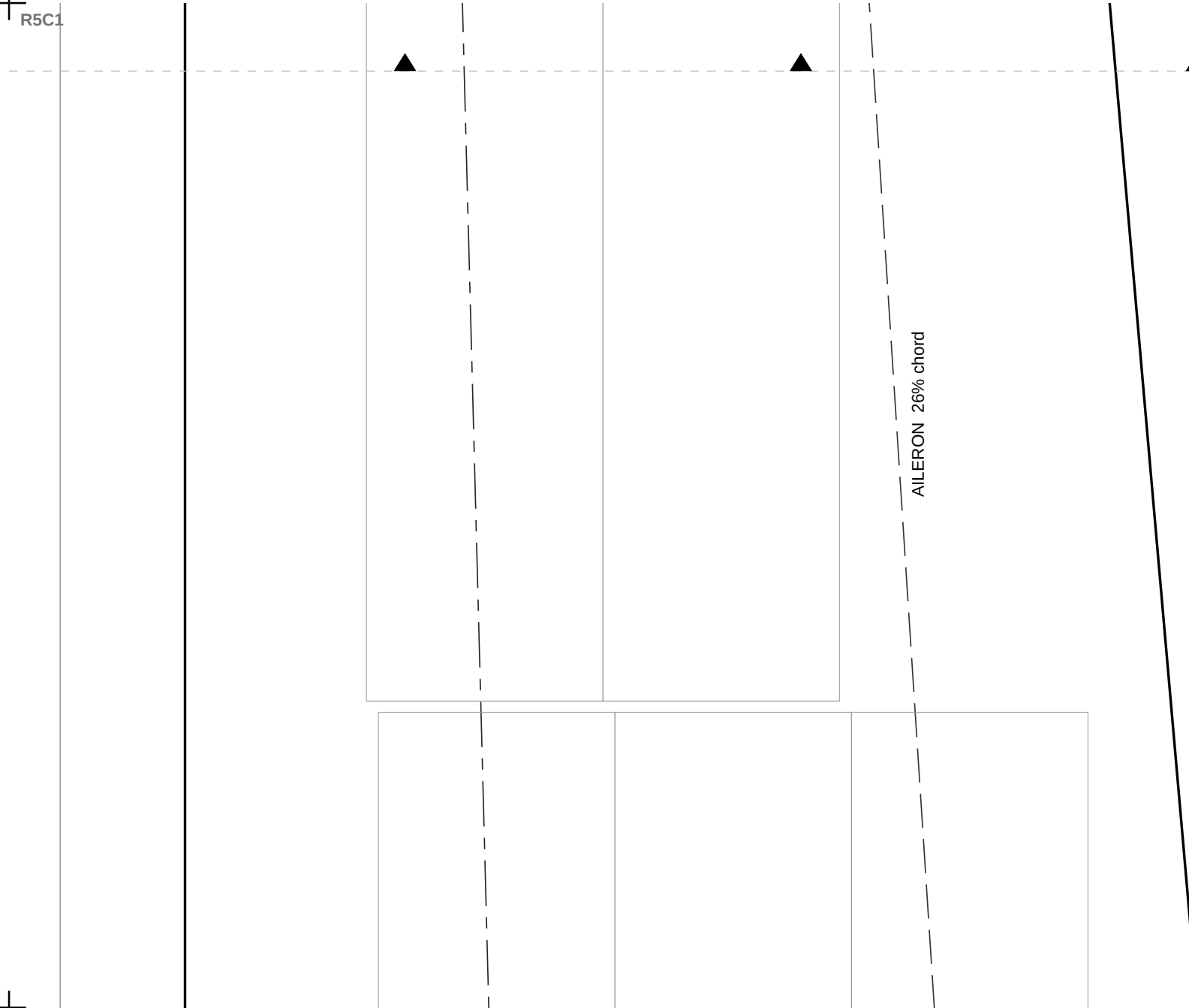
tile row 4 of 7, column 1 of 1 - overlap 12 mm, align the triangles



**SHEET 6 / 22**



R5C1



# WING PANEL

tile row 5 of 7, column 1 of 1 - overlap 12 mm, align the triangles

0

50 mm

100 mm

CHECK BEFORE CUTTING

SHEET 7 / 22

FF-1 Ember - print at 100%





R6C1

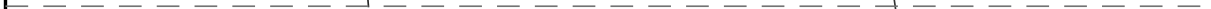


y=960 c=128



faint boxes: solar cell strips on the  
composite version -- ignore for foam

y=850 c=160



## WING PANEL

tile row 6 of 7, column 1 of 1 - overlap 12 mm, align the triangles

0



50 mm

100 mm

CHECK BEFORE CUTTING

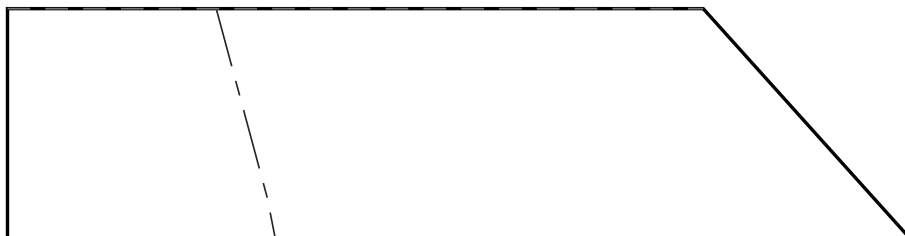


SHEET 8 / 22

FF-1 Ember - print at 100%

TIP 92

y=1000 c=92



**WING PANEL**

tile row 7 of 7, column 1 of 1 - overlap 12 mm, align the triangles

0

50 mm

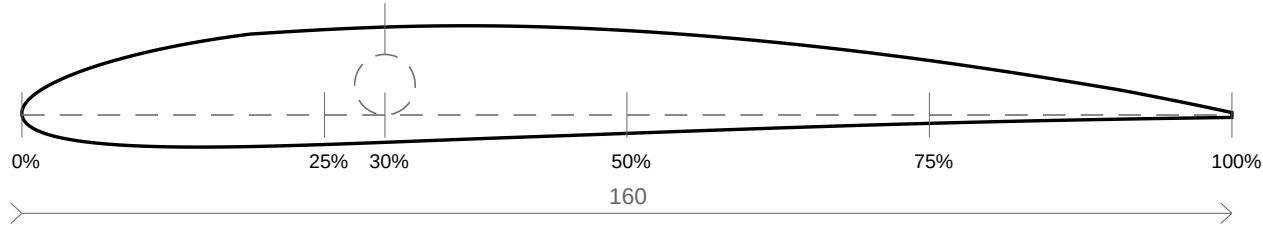
100 mm

CHECK BEFORE CUTTING

## STATION y=850

chord 160 mm

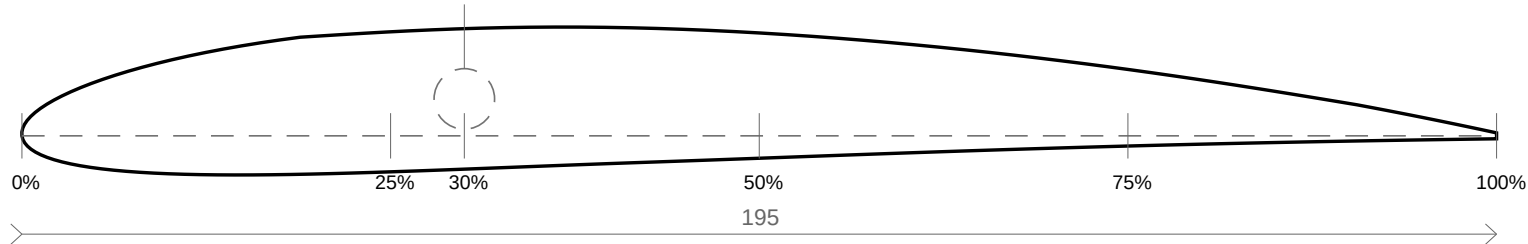
spar 8 mm (section 15.2 mm here)



## ROOT + PANEL BREAK (cut 2)

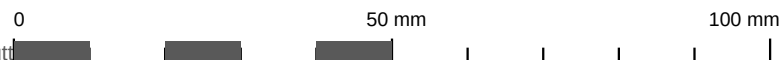
chord 195 mm

spar 8 mm (section 18.6 mm here)

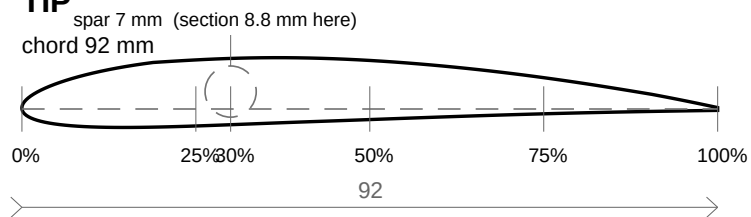


## AEROFOIL TEMPLATES

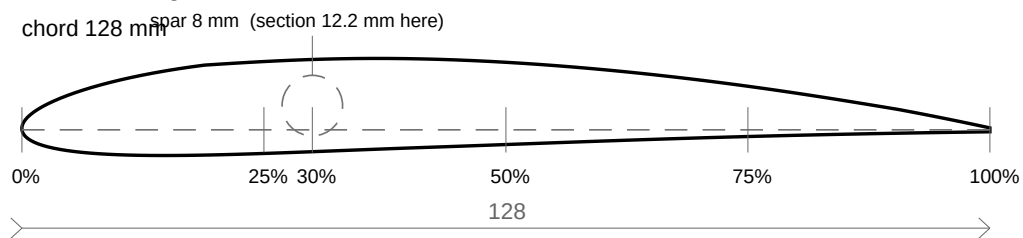
FF-SC1 at full size - transfer to 1.5 mm ply or aluminium for hot-wire cutting



## TIP

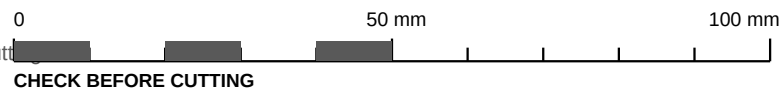


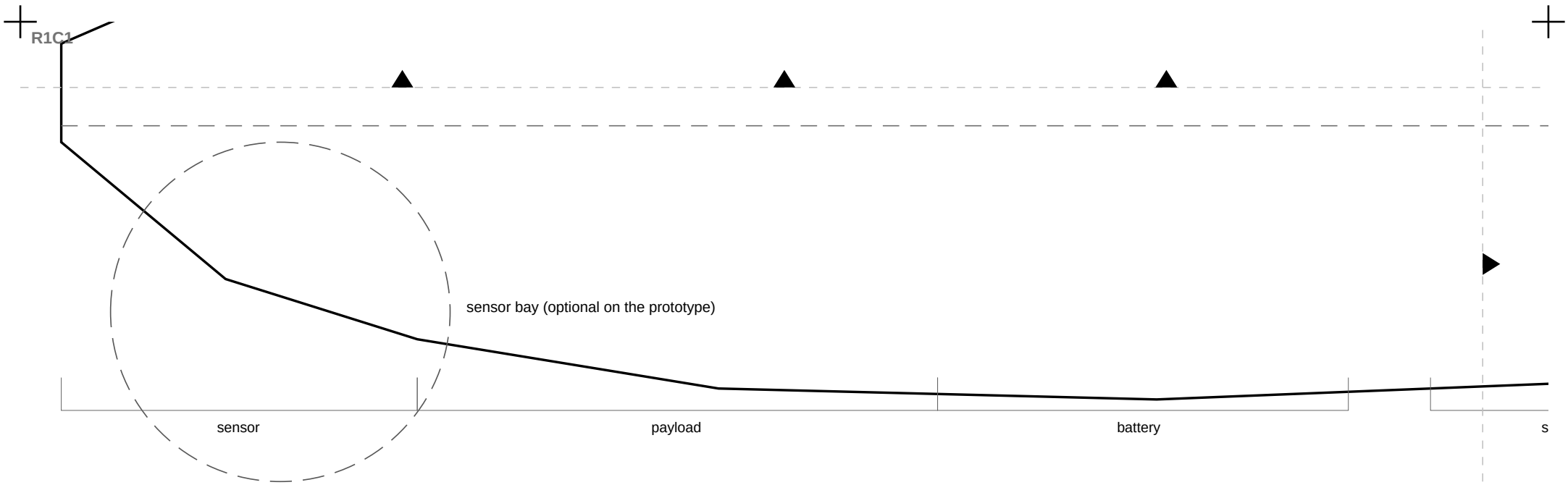
## STATION y=960



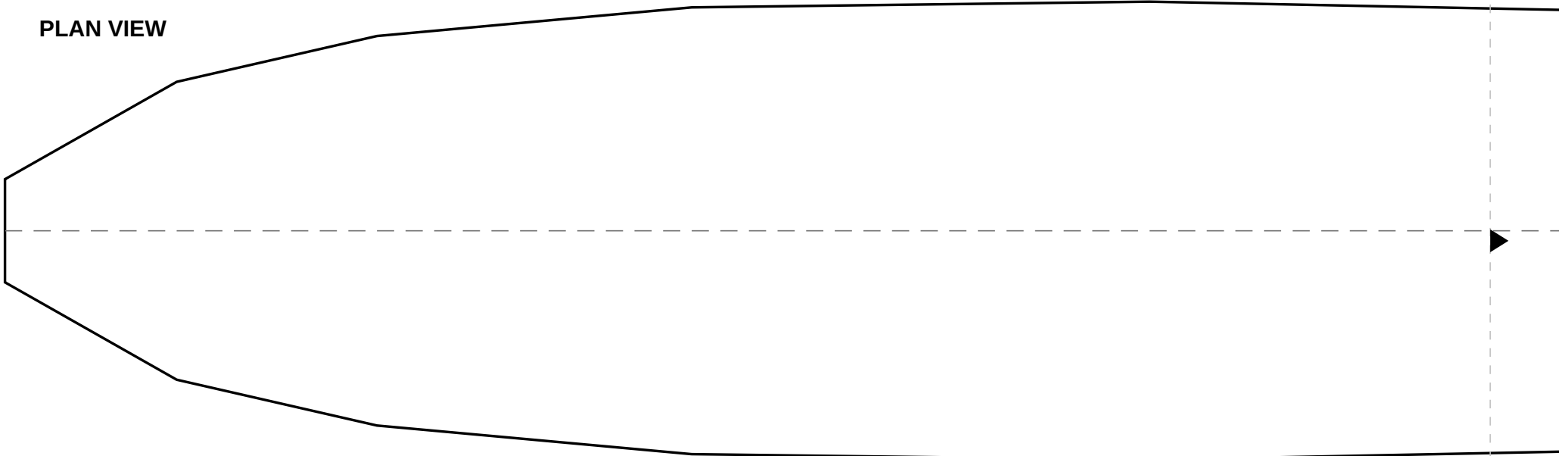
## AEROFOIL TEMPLATES

FF-SC1 at full size - transfer to 1.5 mm ply or aluminium for hot-wire cutting



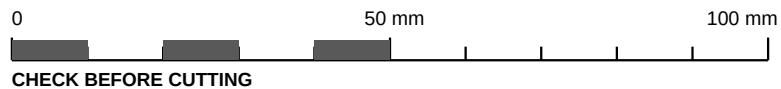


## PLAN VIEW



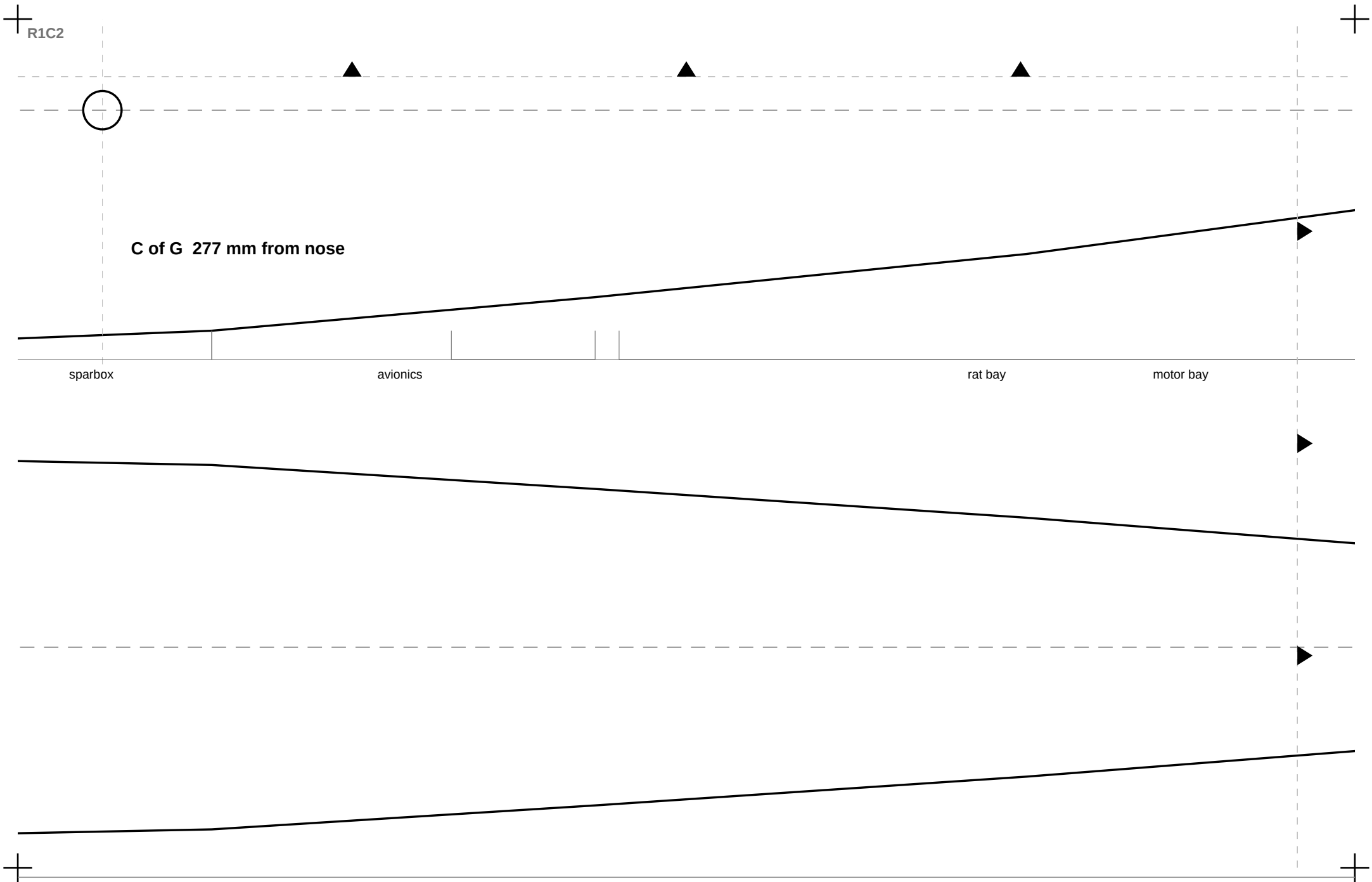
## FUSELAGE POD

tile row 1 of 2, column 1 of 3 - overlap 12 mm, align the triangles



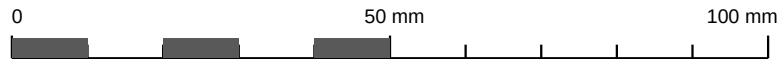
SHEET 12 / 22

FF-1 Ember - print at 100%



**FUSELAGE POD**

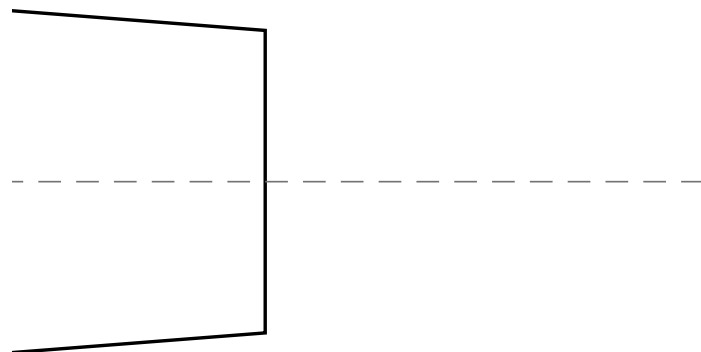
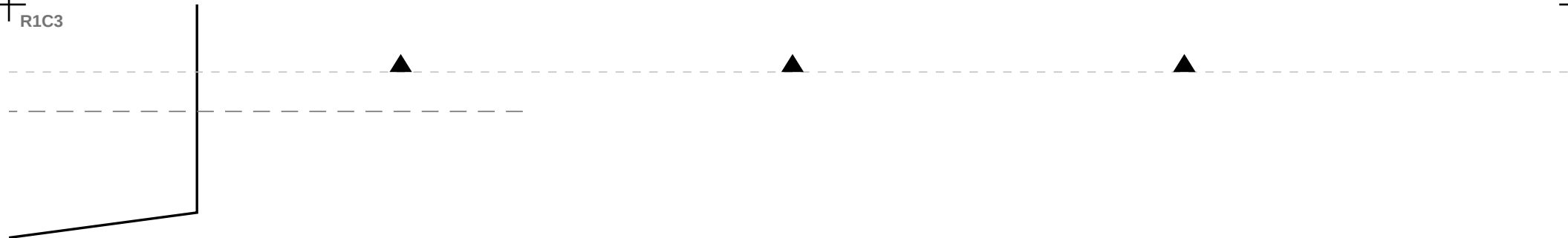
tile row 1 of 2, column 2 of 3 - overlap 12 mm, align the triangles



**CHECK BEFORE CUTTING**

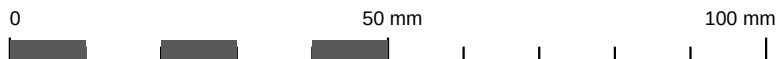


R1C3



# FUSELAGE POD

tile row 1 of 2, column 3 of 3 - overlap 12 mm, align the triangles



CHECK BEFORE CUTTING





## SIDE VIEW - cut 2 sides from 3 mm depron

boom socket at x=620: 16 mm carbon tube, 660 mm long (90 mm buried in the pod)

WING SEAT LE 215, TE 410, incidence



### FUSELAGE POD

tile row 2 of 2, column 1 of 3 - overlap 12 mm, align the triangles

0

50 mm

100 mm

CHECK BEFORE CUTTING

SHEET 15 / 22

FF-1 Ember - print at 100%

+  
R2C2

+

POD LENGTH 620

ncidence +1.5 deg

+

+

## FUSELAGE POD

tile row 2 of 2, column 2 of 3 - overlap 12 mm, align the triangles

0

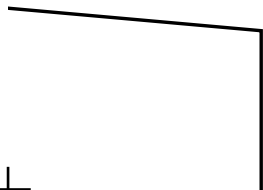
50 mm

100 mm

CHECK BEFORE CUTTING

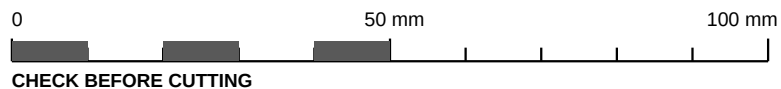
SHEET 16 / 22

FF-1 Ember - print at 100%



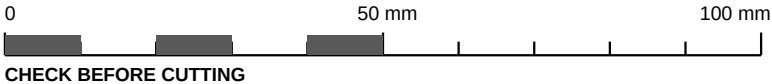
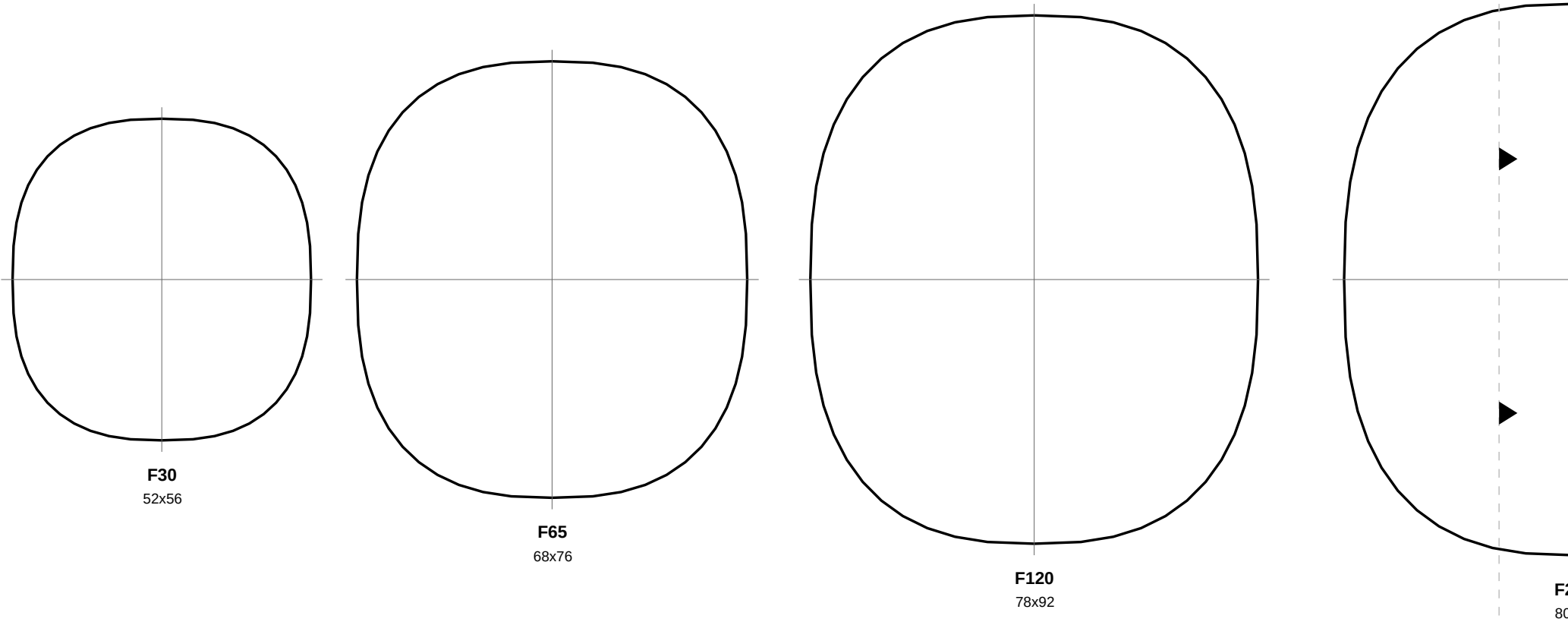
## FUSELAGE POD

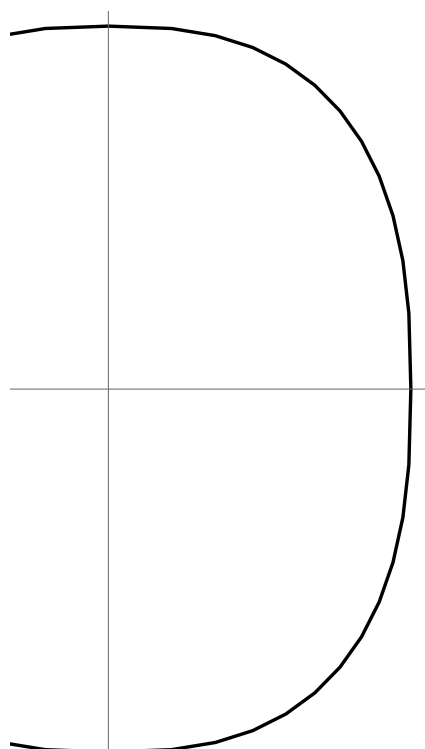
tile row 2 of 2, column 3 of 3 - overlap 12 mm, align the triangles



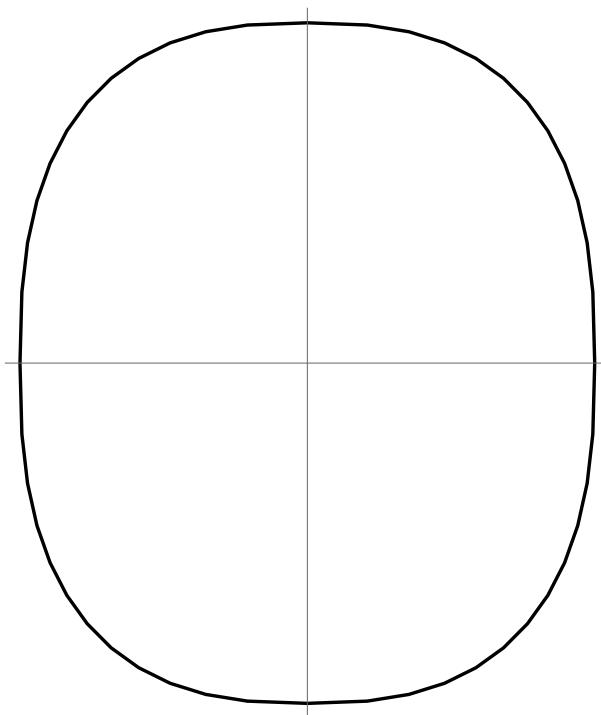
**FORMERS - cut from 3 mm ply or 5 mm depron**

F<number> is the station in mm from the nose; the horizontal tick is the fuselage datum

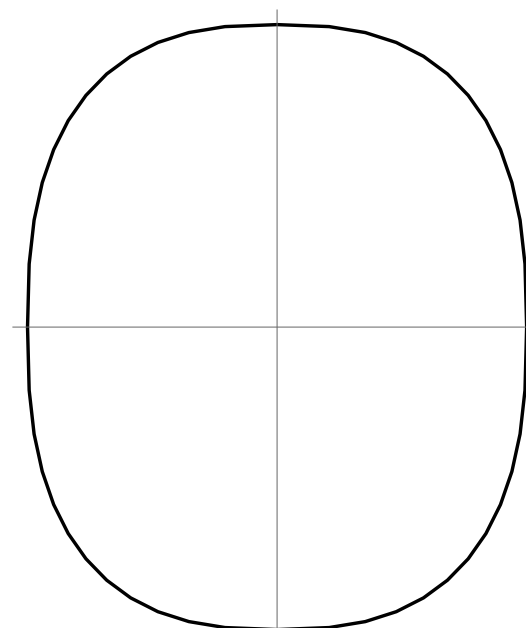




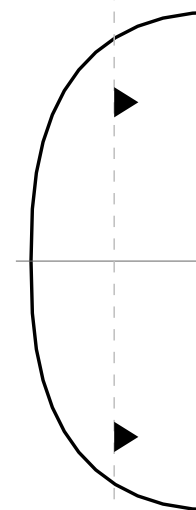
**F200**  
80x96

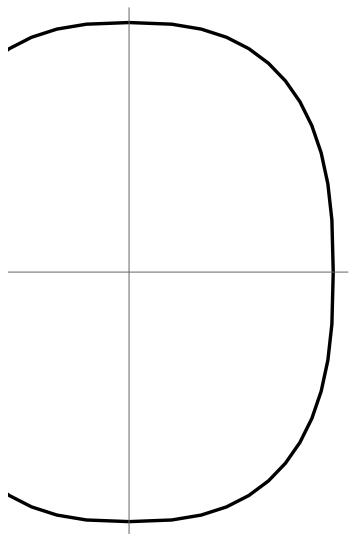


**F300**  
76x90

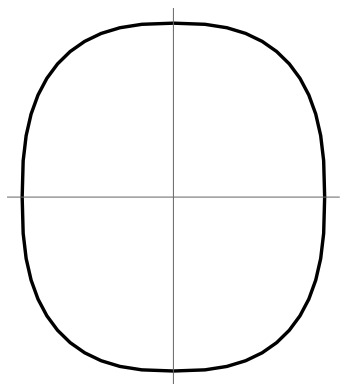


**F380**  
66x80





**F470**  
54x66

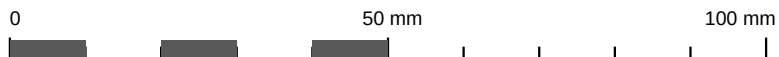


**F560**  
40x46



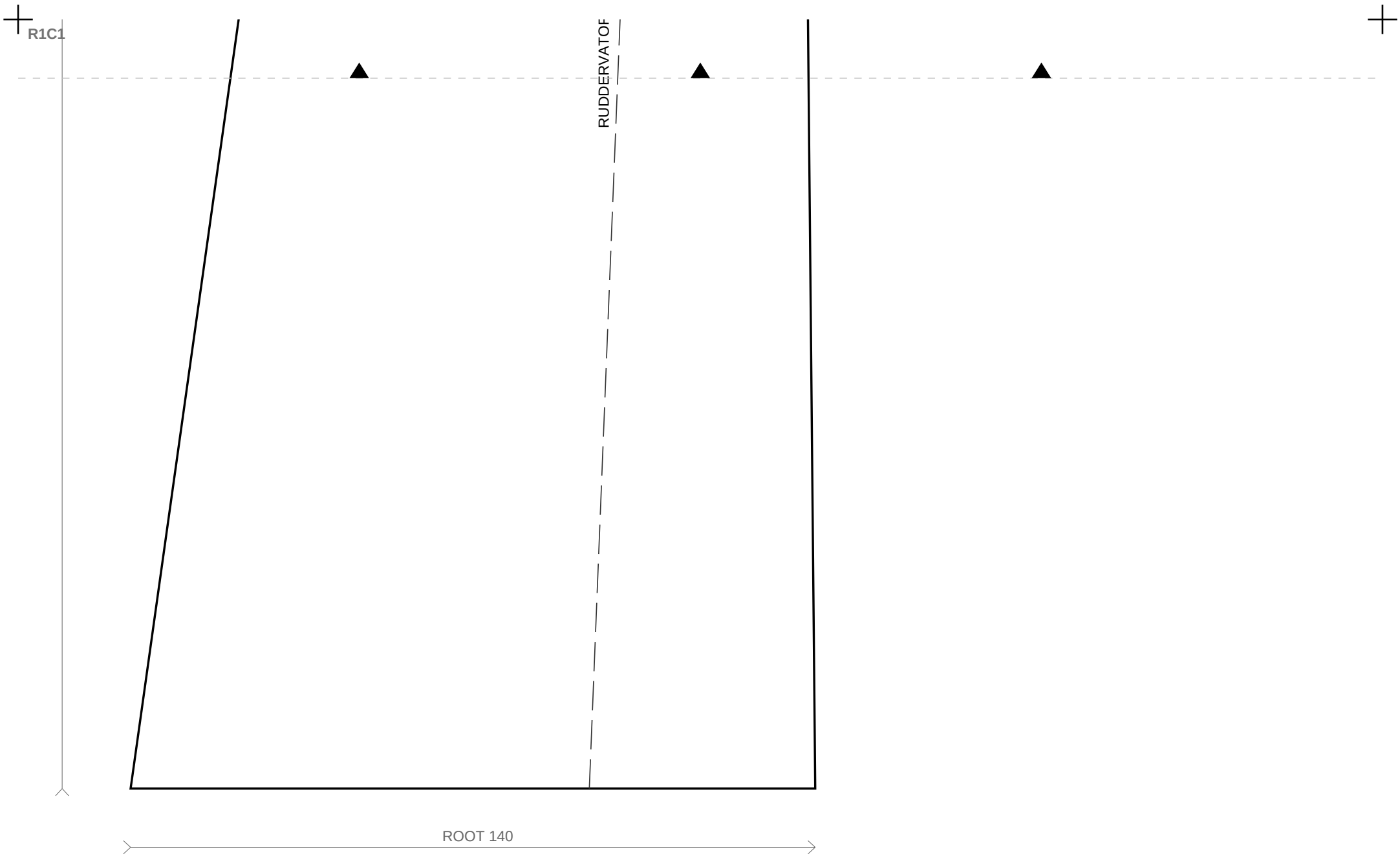
## FORMERS

tile row 1 of 1, column 3 of 3 - overlap 12 mm, align the triangles



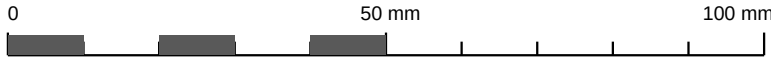
CHECK BEFORE CUTTING





**V-TAIL PANEL**

tile row 1 of 2, column 1 of 1 - overlap 12 mm, align the triangles



CHECK BEFORE CUTTING

## V-TAIL - CUT 2, ONE MIRRORED

dihedral 38 deg from horizontal (76 deg included)

incidence -1.0 deg (decalage 2.5 deg nose-up on the wing)

symmetric section, 9% thick - or flat 5 mm foam with a rounded LE

:RVATOR 33% chord

### V-TAIL PANEL

tile row 2 of 2, column 1 of 1 - overlap 12 mm, align the triangles

0

50 mm

100 mm

CHECK BEFORE CUTTING

### SHEET 22 / 22

FF-1 Ember - print at 100%